CEN-348 Artificial Intelligence Systems
Spring 2021
Midterm Exam
Duration: 180 minutes

Note: Write your name, surname and school number to the left top of your paper. Please write your answers **step by step**. **Draw a box** around your final answers. Please check your answers **carefully**. Upload your answers as a **pdf** file. **Please do not cheat! Please do not share your answers with others.**

1) **(30 points)** Given the following short movie reviews, each labeled with a genre, either **comedy** or **action**:

   1. fun, couple, love, love **comedy**
   2. fast, furious, shoot **action**
   3. couple, fly, fast, fun, fun **comedy**
   4. furious, shoot, shoot, fun **action**
   5. fly, fast, shoot, love **action**

   and a new document **D**:

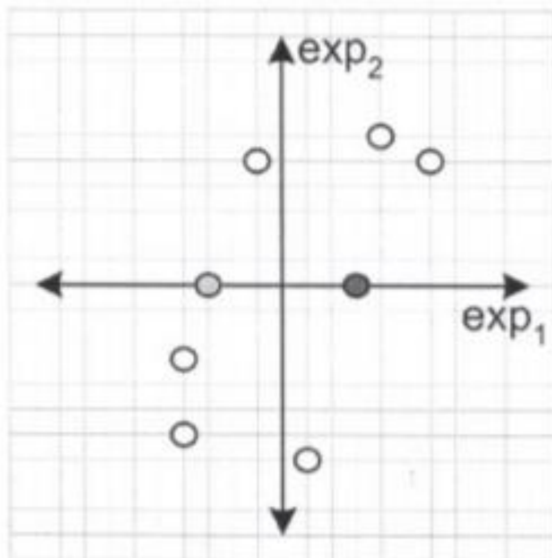
   fast, couple, shoot, fly

   compute the most likely class for **D**. Assume a **Naive Bayes** classifier and use **add-1** smoothing for the likelihoods.

2) **(35 points)** Consider the following gene expression values from 2 microarray experiments for 8 genes.

	exp_1	exp_2
$gene_1$	-4	-3
$gene_2$	6	5
$gene_3$	1	-7
$gene_4$	-4	-6
$gene_5$	4	6
$gene_6$	-1	5
$gene_7$	-3	0
$gene_8$	3	0

Negative expression values in the table mean that the gene is downregulated (i.e., expressed less) in the experimented cell, and positive values mean that the gene is upregulated (i.e., overexpressed). A biologist is trying to find out whether these 8 genes can be separated into two groups based on their behavior in the experimented conditions. In order to visualize the relationships, she sketched a 2-dimensional plot of the genes shown below.



Use k-means clustering to cluster these 8 genes into 2 clusters. Use *gene*$_7$ (light grey data point on the 2-d plot) as the initial cluster center for *cluster*$_l$ and use *gene*$_8$ (dark grey data point) as the initial cluster center for *cluster*$_2$. Indicate which data point belongs to which cluster and also give the coordinates of the centroids at each iteration. Iterate until convergence, i.e., until no change in clusters. When computing distances to centroids, you may use the squared distance (no need to take the square-root).

3) **(35 points)** Suppose you want to train a **linear regression** model $y = \beta_1 x + \beta_2$ using the training data below. According to this, compute β_1 and β_2 iteratively by using batch gradient descent. Write the new values of β_1 and β_2 at the end of each iteration.

x	y
1	17
2	15
3	10
4	2